Reg. No. | | | | | | | | | | | [12AE832]

EIGHTH SEMESTER B.E UNIVERSITY EXAMINATION, JUNE – 2016

VIBRATION CONTROL

Time: 3 Hours **Maximum Marks:70**

Instructions:

1. Answer any **FIVE** full Questions.
2. Missing data may be suitably assumed.

1.	**a.**	Explain the term mechanical vibration.	**04**
	b.	What are the sources of vibration in a rotatory machine?	**05**
	c.	Explain self-excited vibrations due to exposure to wind.	**05**
2.	**a.**	Explain the equation of motion of a spring, mass, viscous damping system which is acted upon by dynamic force.	**05**
	b.	Derive expansion for energy absorbed per cycle by viscous damper.	**04**
	c.	Explain transmissibility and how it is influenced by damping coefficient.	**05**
3.	**a.**	Describe typical high vibration work environment.	**05**
	b.	What are the harmful effects of vibration on human beings?	**04**
	c.	Explain the effects of vibration on aircraft and helicopter.	**05**
4.	**a.**	Explain the construction of a viscous fluid damper.	**04**
	b.	Describe friction damping.	**04**
	c.	A vibrating system with a friction damping is subjected to cyclic displacements, $x=100 \sin\omega t$ mm where $\omega=31.4$ rad/sec. Damping force is 20N. Computer the energy absorbed by the damper per second.	**06**
5.	**a.**	Explain static balancing of a thin disc about the axis.	**05**
	b.	Explain dynamic balancing of a rotor using pivoted carriage balancing machine.	**06**
	c.	What is the need for dynamic balancing after static balancing?	**03**

6. a. Explain the principle and application of air suspension. 06

b. Compute the stiffness of the air spring shown in figure-6b. It is supporting a load=4,000N. 08

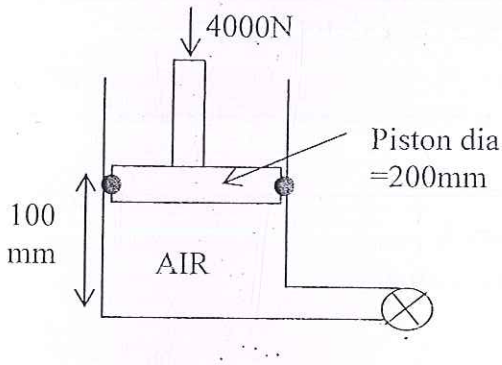

Fig.6b

7. a. Explain the principles of vibration control. 05

b. Explain fluidlastic vibration isolation devices. 05

c. Explain Kelvin linear model for viscoelastic material. 04

8. Write short notes on, the following:

a. Pendulum absorber on rotor blade 3.5

b. Hysteresis in metallic materials 3.5

c. Structural detuning 3.5

d. Elastomeric bushes. 3.5